

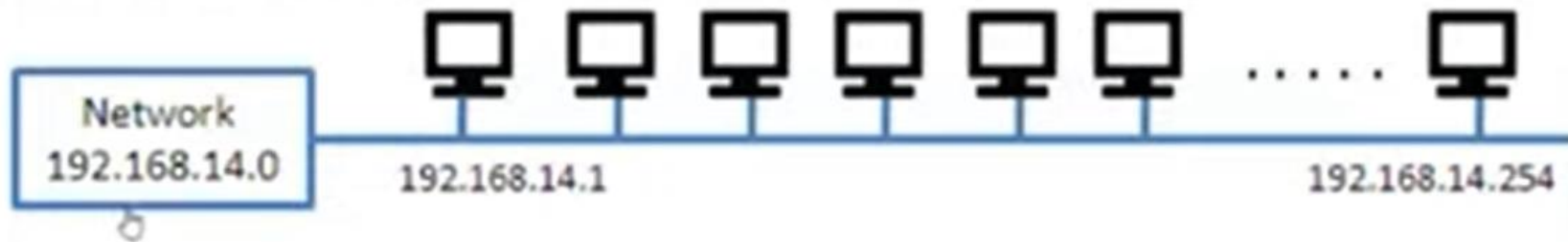
Subnetting and Supernetting

- During the era of classful addressing, subnetting was introduced.
- If an organization was granted a large block in class A or B, it could divide the addresses into several contiguous groups and assign each group to smaller networks called **subnets**
- The practice of dividing a network into two or more networks is called **subnetting**

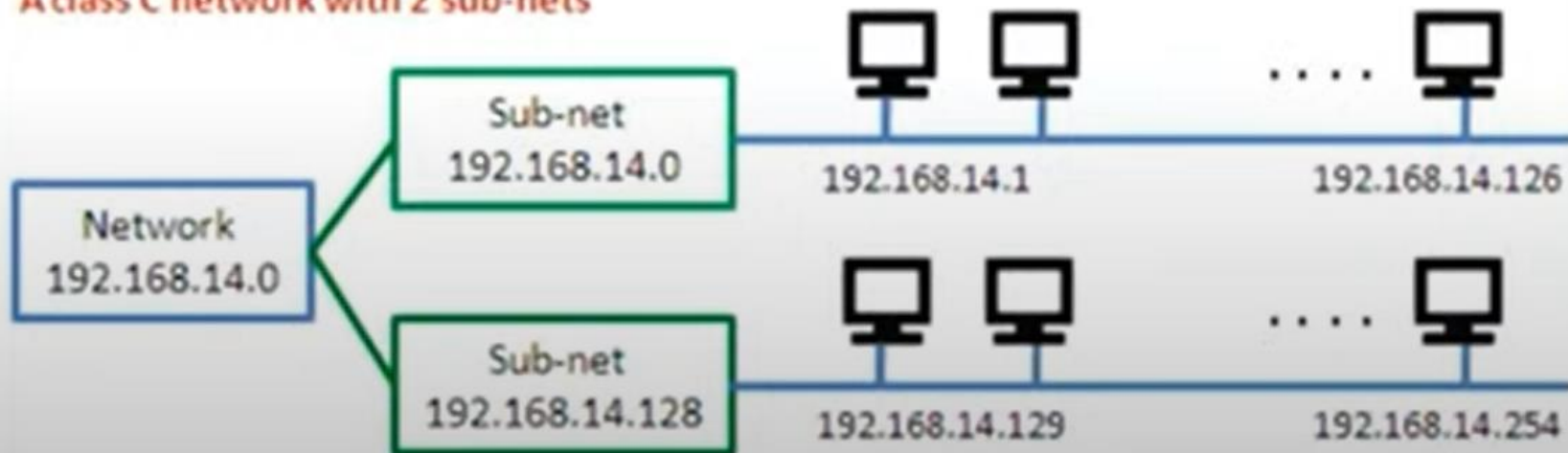
Subnetting and Supernetting

IP Sub-netting Example

A class C network without sub-netting

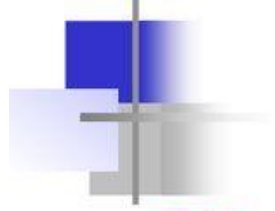


A class C network with 2 sub-nets



Supernetting

- The time came when most of the class A and class B addresses were depleted; however, there was still a huge demand for midsize blocks.
- The size of a class C block with a maximum number of 256 addresses did not satisfy the needs of most organizations.
- Even a midsize organization needed more addresses. One solution was supernetting.
- In supernetting, an organization can combine several class C blocks to create a larger range of addresses.
- In other words, several networks are combined to create a supernetwork or a supernet.
- An organization can apply for a set of class C blocks instead of just one. For example, an organization that needs 1000 addresses can be granted four contiguous class C blocks. The organization can then use these addresses to create one supernetwork. Supernetting decreases the number of 1s in the mask. For example, if an organization is given four class C addresses, the mask changes from /24 to /22. We will see that classless addressing eliminated the need for supernetting

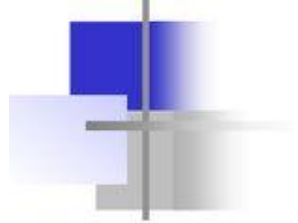


Network Addresses

The network address is the first address.

The network address defines the network to the rest of the Internet.

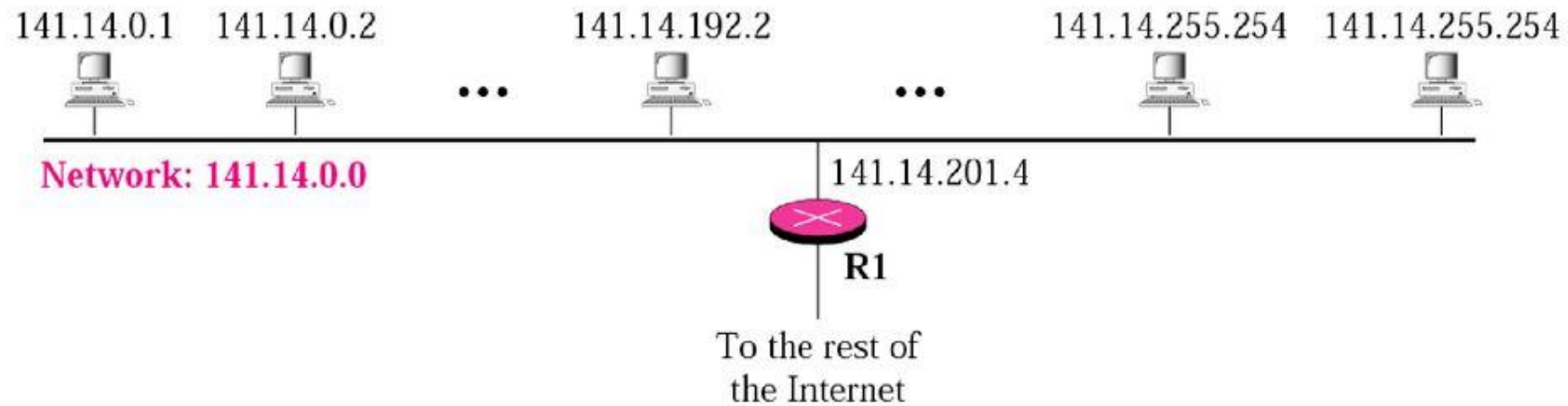
Given the network address, we can find the class of the address, the block, and the range of the addresses in the block

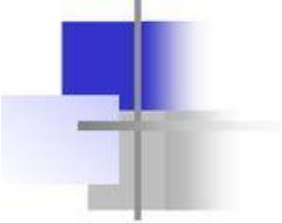


IP-Addressing/Subnetting

- a) IP address designed with 2 levels of hierarchy: network-ID & host-ID.
- b) However, often organisation needs to assemble the hosts into groups: the network needs to be divided into several subnetworks (subnets); hence requires 3 levels of hierarchy. (netid: subnetid : hostid)
- c) Remember that, the outside world only knows the organisation by its **network address**. Inside the organisation each sub-network is recognised by its **sub-network address**.
- d) In subnetting, a network is divided into several smaller groups that have its own subnet address depends on the hierarchy of subnetting but still appear as a single network to the rest of the Internet.
- e) The question is how a router knows whether it is a network address or a subnet? The key is using the **subnet mask**. (similar to def. mask).
- f) Only the network administrator knows about the network address and subnet address but router does not. External router has routing table based on network addresses; Internal router has routing table based on subnetwork addresses.

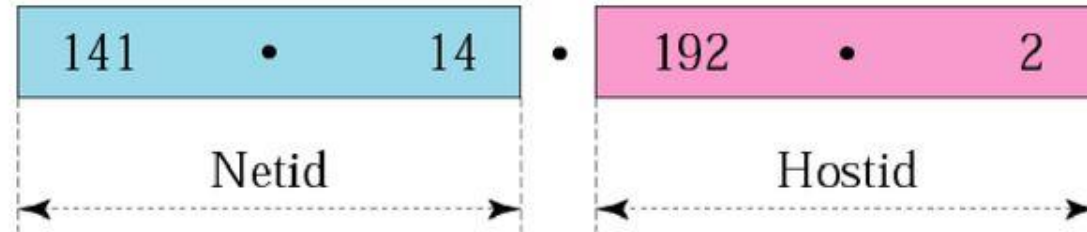
A network with two levels of hierarchy (not subnetted)



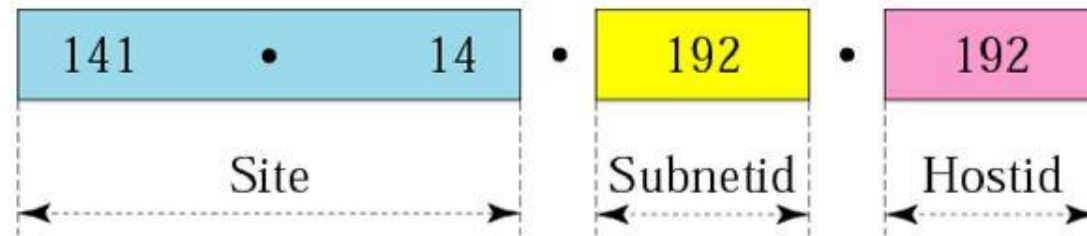


Addresses in a network

With and without subnetting



a. Without subnetting



b. With subnetting

Just like telephone system

(408) 864 – 8902

Area code Exchange Connection

Network Addresses/subnetting

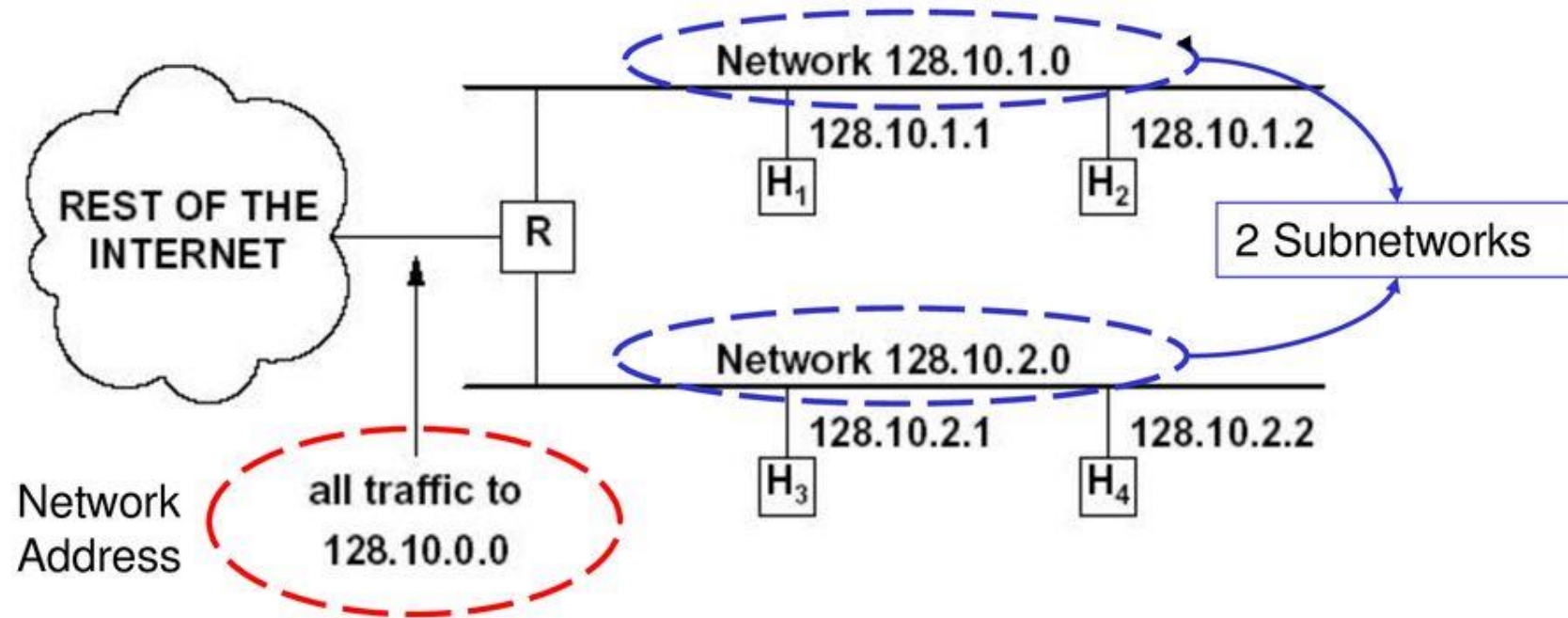
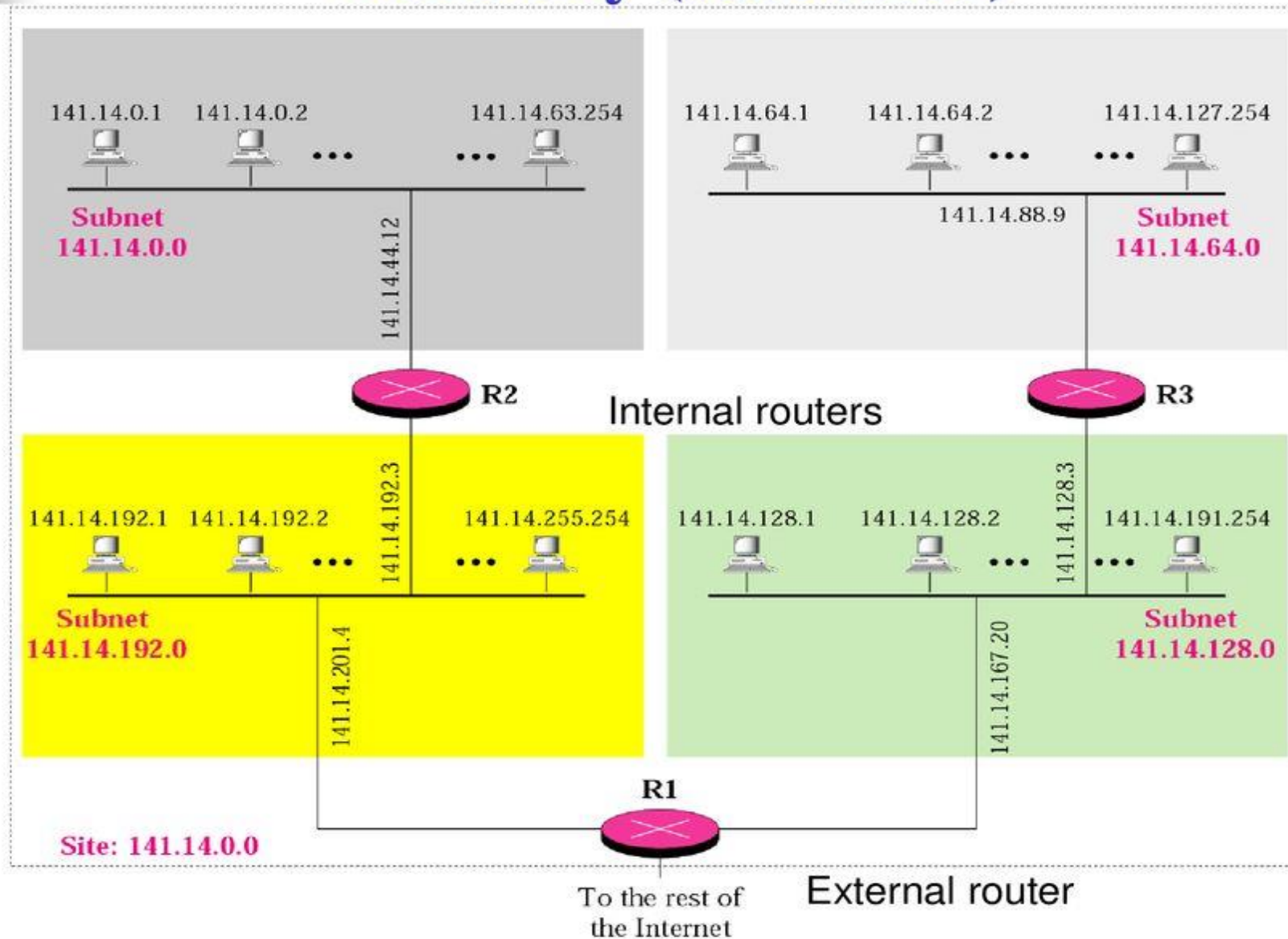
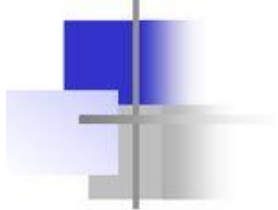


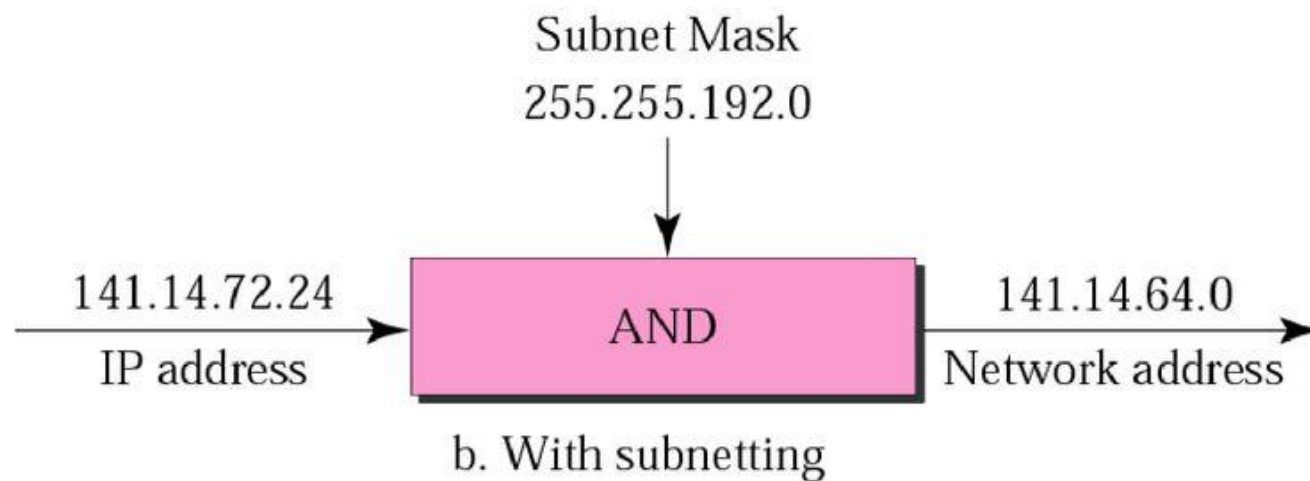
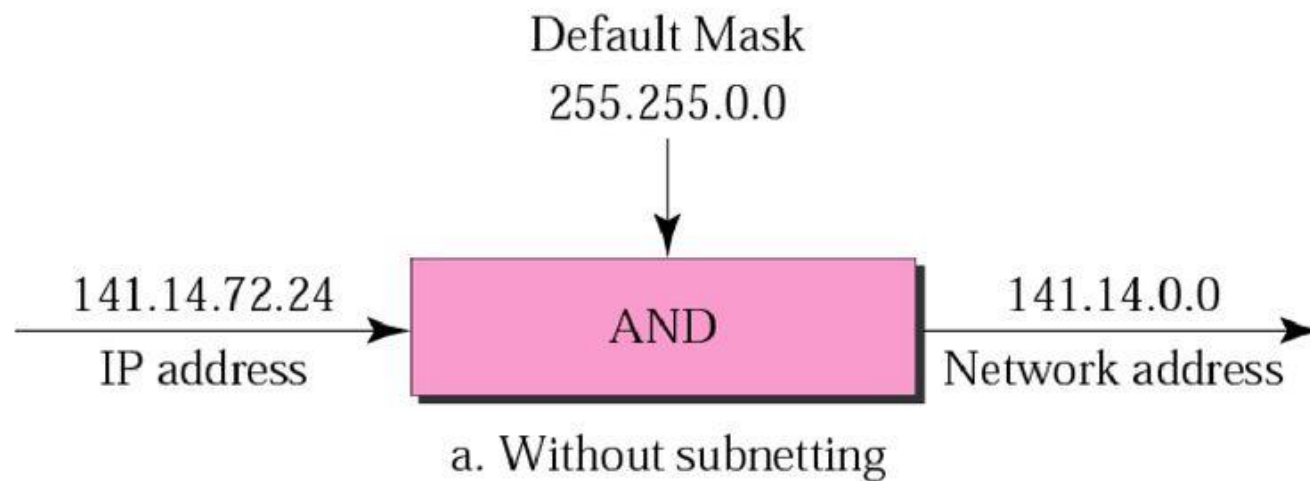
Figure 10.3 A site with two physical networks using subnet addressing to label them with a single class *B* network address. Router *R* accepts all traffic for net 128.10.0.0 and chooses a physical network based on the third octet of the address.

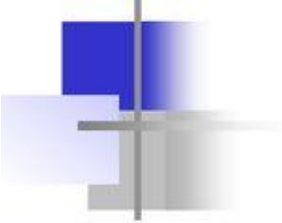
A network with three levels of hierarchy (subnetted)





Default mask and subnet mask



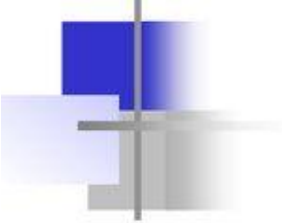


Finding the Subnet Address

Given an IP address, we can find the **subnet address** the same way we found the **network address** in the previous chapter.

We apply the **mask** to the address.

We can do this in two ways:
straight or short-cut.



Straight Method

In the straight method, we use binary notation for both the address and the mask and then apply the AND operation to find the subnet address.

Short-Cut Method

- ** If the byte in the mask is 255, copy the byte in the address.
- ** If the byte in the mask is 0, replace the byte in the address with 0.
- ** If the byte in the mask is neither 255 nor 0, we write the mask and the address in binary and apply the AND operation.

Example

What is the subnetwork address if the destination address is 200.45.34.56 given that the subnet mask is 255.255.240.0?

Solution

11001000 00101101 00100010 00111000

11111111 11111111 11110000 00000000

11001000 00101101 00100000 00000000

The subnetwork address is **200.45.32.0**.



Example

What is the subnetwork address if the destination address is 19.30.84.5 and the mask is 255.255.192.0?

Example

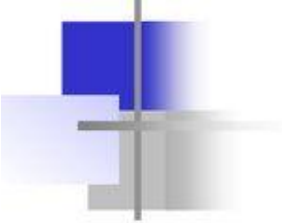
What is the subnetwork address if the destination address is 19.30.84.5 and the mask is 255.255.192.0?

Solution

Answer: Subnet Address = 19.30.64.0

IP Address									
19	•	30	•	84	•	5			
Mask									
255	•	255	•	192	•	0			
19	•	30	•	64	•	0			
Subnet Address									

84	0	1	0	1	0	1	0	0
192	1	1	0	0	0	0	0	0
64	0	1	0	0	0	0	0	0



Comparison of a default mask and a subnet mask

	255.255.0.0			
Default Mask	11111111 11111111		00000000 00000000	
			16	
	255.255.224.0			
Subnet Mask	11111111 11111111		111	00000 00000000
			3	13

Note

The number of subnets must be a power of 2.



Example

A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.

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Solution

The number of 1s in the default mask is 24 (class C).

The company needs six subnets. Since 6 is not a power of 2, the next number that is a power of 2 is 8 (2^3). That means up to 8 subnets.

Hence, we need 3 more '1's in the subnet mask =
11111111.11111111.11111111.11100000 or 255.255.255.224

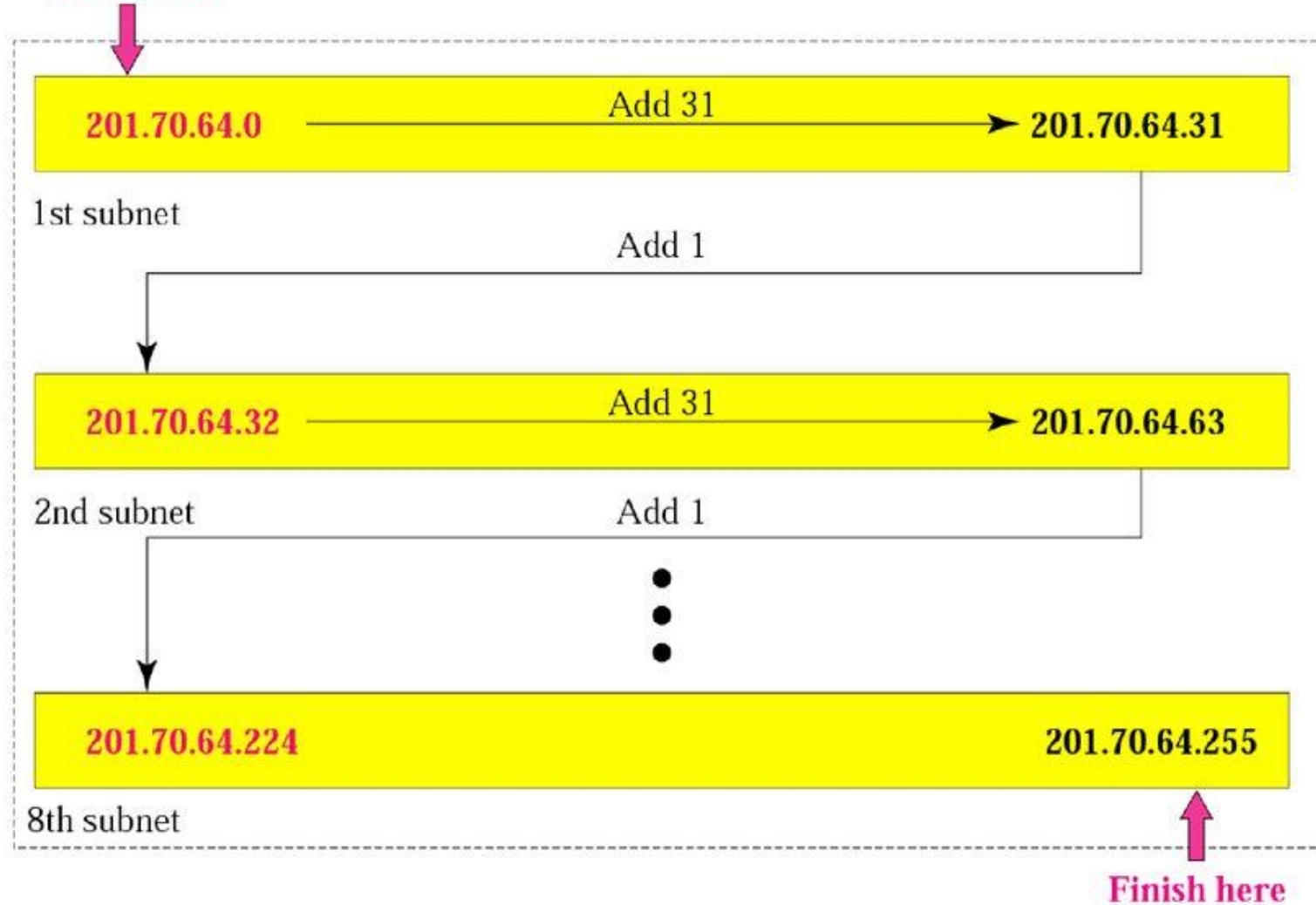
The total number of 1s in the subnet mask is 27 ($24 + 3$).

Since the total number of 0s is 5 ($32 - 27$).

The number of addresses in each subnet is 2^5
(5 is the number of 0s) or 32.

Solution (Continued)

Start here





Example

A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Design the subnets.

Example

A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Design the subnets.

Solution

The number of 1s in the default mask is 16 (class B).

The company needs 1000 subnets. Since it is not a power of 2, the next number is 1024 (2^{10}). We need 10 more 1s in the subnet mask.

The total number of 1s in the subnet mask is 26 ($16 + 10$).

The total number of 0s is 6 ($32 - 26$).

Solution (Continued)

The submask is

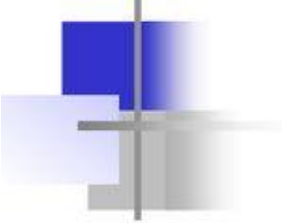
11111111 11111111 11111111 110000000

or

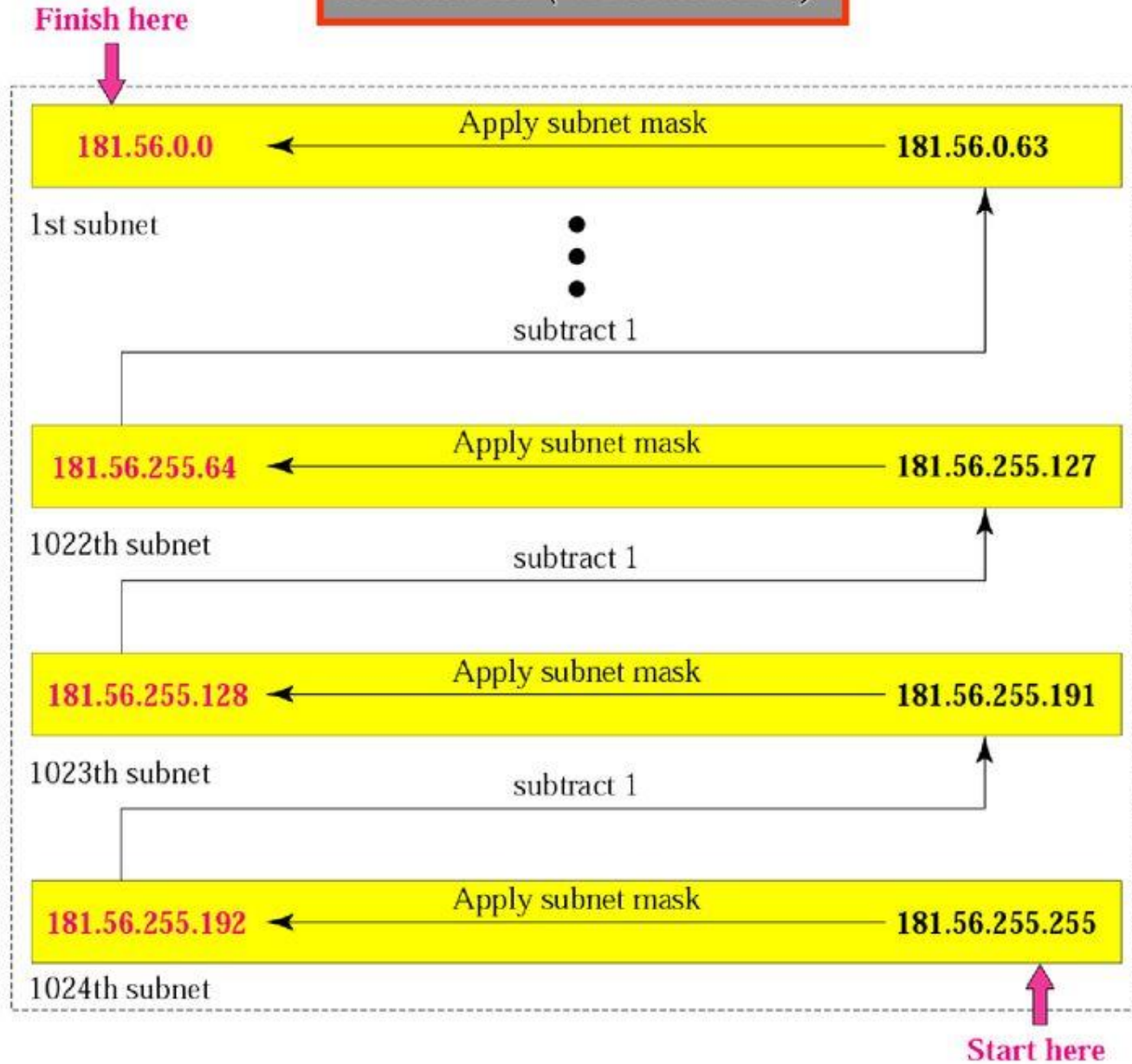
255.255.255.192.

The number of subnets is 1024.

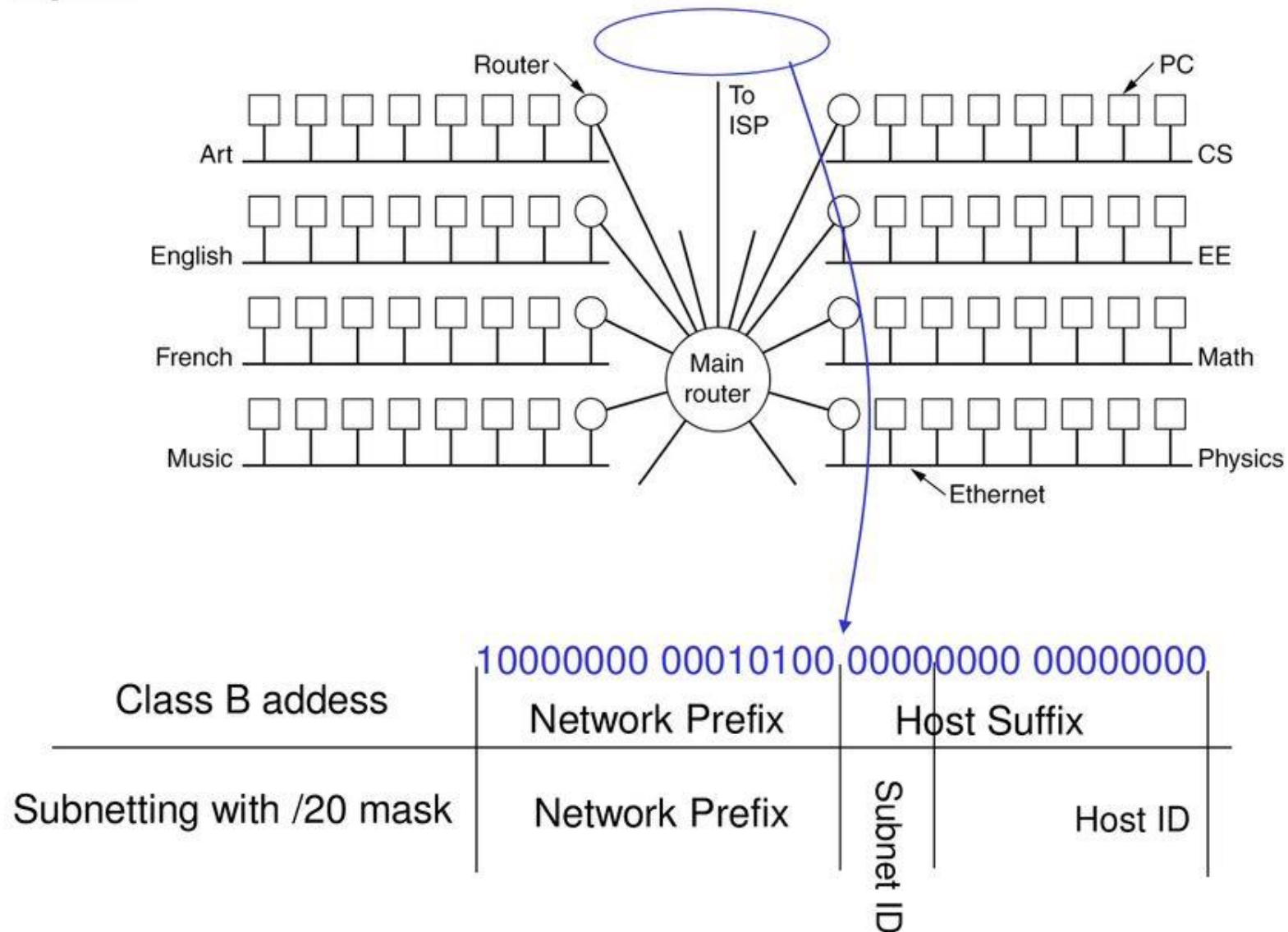
The number of addresses in each subnet is 2^6
(6 is the number of 0s) or 64.



Solution (Continued)

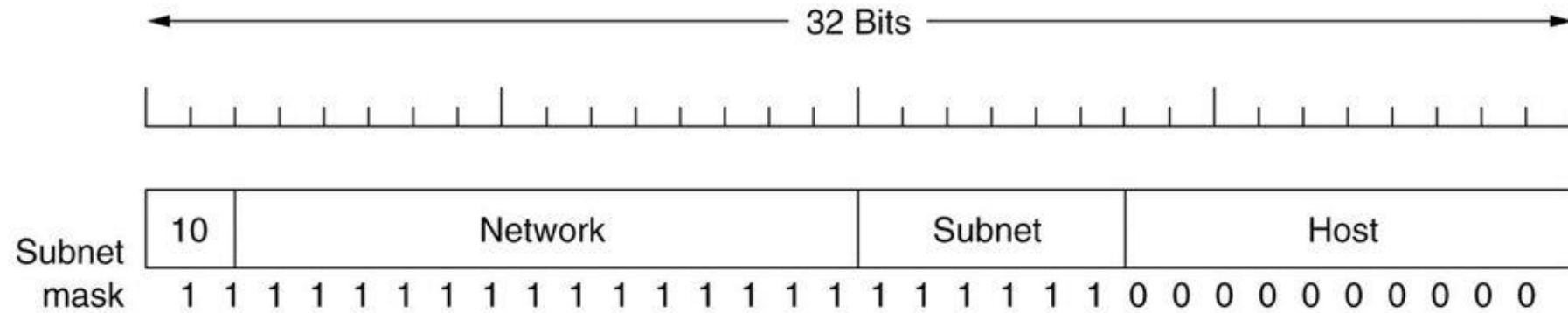


Subnetting in Classful Addresses





Subnetting in Classful Addresses





Example

An organization is granted the network address block of 130.34.12.64/26. The organization needs to have four subnets. What are the subnet addresses and their range for each subnet?

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An organization is granted the network address block of 130.34.12.64/26. The organization needs to have four subnets. What are the subnet addresses and their range for each subnet?

Solution

The suffix length is 6 (32-26). This means the total number of addresses in the block is 64 (2^6). If we create four subnets, each subnet will have 16 addresses. Let us first find the subnet prefix (subnet mask). We need four subnets, which means we need to add two more '1's to the site prefix /26. The subnet prefix is then /28.

Subnet 1: 130.34.12.64/28 to 130.34.12.79/28.

Subnet 2 : 130.34.12.80/28 to 130.34.12.95/28.

Subnet 3: 130.34.12.96/28 to 130.34.12.111/28.

Subnet 4: 130.34.12.112/28 to 130.34.12.127/28.

Subnet the IP address 216.21.5.0 into 30 hosts in each subnet.

SUBNETTING – 5 STEPS

1. Identify the class of the IP address and note the Default Subnet Mask.
2. Convert the Default Subnet Mask into Binary.
3. Note the number of hosts required per subnet and find the Subnet Generator (SG) and octet position.
4. Generate the new subnet mask.
5. Use the SG and generate the network ranges (subnets) in the appropriate octet position.

1. Class C – Default Subnet Mask: 255.255.255.0

2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0

3. No. of hosts/subnet: 30 (11110) – 5 bits SG: 32 Octet Position: 4

1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 0 0 0 0 0

4. New subnet mask: 255.255.255.224 or /27

5. Network Ranges (Subnets)

216.21.5.0

216.21.5.32

216.21.5.64

216.21.5.96

216.21.5.128

QUESTION

Subnet the IP address 150.15.0.0 into 500 hosts in each subnet.



1. Class B – Default Subnet Mask: 255.255.0.0

2. 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0

3. No. of hosts/subnet: 500(111110100)–9 bits SG: 2 Octet Position: 3

1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 1 . 1 1 1 1 1 1 1 0 . 0 0 0 0 0 0 0 0

4. New subnet mask: 255.255.254.0 or /23

5. Network Ranges (Subnets)

150.15.0.0 – 150.15.1.255

150.15.2.0

150.15.4.0

150.15.6.0

150.15.8.0

and so on....

An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be

An organization has a class B network and wishes to form subnets for 64 departments. The subnet mask would be

(A) 255.255.0.0

(B) 255.255.64.0

(C) 255.255.128.0

(D) 255.255.252.0 ✓

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- a. The first group has 64 customers; each needs 256 addresses.*
- b. The second group has 128 customers; each needs 128 addresses.*
- c. The third group has 128 customers; each needs 64 addresses.*

Design the subblocks and find out how many addresses are still available after these allocations.

Figure 19.9 shows the situation.

Group 1

For this group, each customer needs 256 addresses. This means that 8 ($\log_2 256$) bits are needed to define each host. The prefix length is then $32 - 8 = 24$. The addresses are

<i>1st Customer:</i>	<i>190.100.0.0/24</i>	<i>190.100.0.255/24</i>
<i>2nd Customer:</i>	<i>190.100.1.0/24</i>	<i>190.100.1.255/24</i>
<i>...</i>		
<i>64th Customer:</i>	<i>190.100.63.0/24</i>	<i>190.100.63.255/24</i>
<i>Total = $64 \times 256 = 16,384$</i>		

Group 2

For this group, each customer needs 128 addresses. This means that 7 ($\log_2 128$) bits are needed to define each host. The prefix length is then $32 - 7 = 25$. The addresses are

1st Customer:	190.100.64.0/25	190.100.64.127/25
2nd Customer:	190.100.64.128/25	190.100.64.255/25
...		
128th Customer:	190.100.127.128/25	190.100.127.255/25
Total = $128 \times 128 = 16,384$		

Group 3

For this group, each customer needs 64 addresses. This means that 6 ($\log_2 64$) bits are needed to each host. The prefix length is then $32 - 6 = 26$. The addresses are

<i>1st Customer:</i>	<i>190.100.128.0/26</i>	<i>190.100.128.63/26</i>
<i>2nd Customer:</i>	<i>190.100.128.64/26</i>	<i>190.100.128.127/26</i>
<i>...</i>		
<i>128th Customer:</i>	<i>190.100.159.192/26</i>	<i>190.100.159.255/26</i>
<i>Total = $128 \times 64 = 8192$</i>		

Number of granted addresses to the ISP: 65,536

Number of allocated addresses by the ISP: 40,960

Number of available addresses: 24,576

ISP

To and from
the Internet

